Hardware Architecture

Software Architecture

Local GUI & Display Driver

Analytical Core

DPV Waveform Generation

Differential Current Calculation

Web & Data

Data Storage: SPIFFS & SD

Embedded Web Server & Wi-Fi

FreeRTOS Task Management

Physical Interface

2x Push Buttons & 1x Rotary Encoder

Onboard Display e.g., TFT/OLED

GPIO Interrupts

ESP32-S3 Microcontroller

SPI

SPI

3.3V Power

Power Management / Battery

SPI Bus & Interrupts

MicroSD Card Module

Isolated Power

AD5941 Analog Front-End SoC

Current Source/Sink

Buffer

Analog Potential

Fluid Sample

Counter Electrode

Reference Electrode

Working Electrode